Perovskite switching: Worked solutions

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

E01 Area scaling with a parallel path

$R_i = RA/A$ gives $10\,\Omega$ at $10\,\mu\mathrm{m}^2$ and $2.5\,\Omega$ at $40\,\mu\mathrm{m}^2$. The total is $R = (R_i^{-1} + 100^{-1})^{-1}$, respectively $9.0909\,\Omega$ and $2.4390\,\Omega$. A roughly inverse-area trend does not prove a unique interface path: the assumed parallel conduction already changes the result, and contacts, current crowding and state variations also matter.

E02 Depletion width

Convert N_D to $10^{24}\,\mathrm{m}^{-3}$. With $\epsilon = 100\epsilon_0$, $W = \sqrt{2(100)(8.854\times10^{-12})(0.5)/(1.602\times10^{-19}\times10^{24})} = 7.434\times10^{-8}\,\mathrm{m} = 74.3\,\mathrm{nm}$. Quadrupling only N_D scales W by $1/2$, to $37.2\,\mathrm{nm}$. Fixed interfacial charge, incomplete ionization, nonuniform doping or a film thinner than the calculated region can invalidate this result; the potential drop need not stay constant as doping changes.

E03 A log–log slope

$\gamma = (V/I)(dI/dV) = (a + 2bV)/(a + bV)$. Substitution gives $\gamma = 1.0909$, 1.5 and 1.9091. The quadratic term dominates at large V without proving that it originates from the SCLC model; contact effects, traps, heating and other transport contributions require separate checks.

E04 Oxygen pressure and charge balance

Mass action gives $[V_O]n^2p_{O_2}^{1/2} = K$ under the stated activity convention. Replacing $[V_O]$ by $n/2$ yields $n^3 \propto p_{O_2}^{-1/2}$, so $n \propto p_{O_2}^{-1/6}$. Increasing pressure by 64 multiplies n by $64^{-1/6} = 1/2$. The electron conductivity $qn\mu$ also halves only if μ remains fixed and that channel dominates. Other defects, carriers, activities or non-equilibrium conditions alter the result.

Lin Cing https://github.com/xinhi6716/resistive-memory-self-study

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E05 SCLC assumptions

From $J = qn\mu E$ and $dE/dx = qn/\epsilon$, obtain $E\,dE/dx = J/(\epsilon\mu)$. With constant J, mobility and permittivity, and ideal $E(0) = 0$, integration gives $E(x) = \sqrt{2Jx/(\epsilon\mu)}$. Integrating from 0 to d gives $V = (2/3)\sqrt{2J/(\epsilon\mu)}d^{3/2}$ and hence $J = 9\epsilon\mu V^2/(8d^3)$. The derivation assumes planar one-dimensional, steady, trap-free unipolar drift, ohmic injection, negligible equilibrium carriers and diffusion, constant parameters and the stated boundary. A sum of linear and quadratic contributions or other field-dependent transport can yield a slope near 2, so slope alone is not proof.

E06 Read current and reliability

$I_{\mathrm{HRS}} = 1\,\mathrm{V}/1\,\mathrm{M\Omega} = 1\,\mu\mathrm{A}$ and $I_{\mathrm{LRS}} = 1\,\mathrm{V}/10\,\mathrm{k\Omega} = 100\,\mu\mathrm{A}$. A threshold requires the state-current distributions, sense noise and an acceptable error rate; retention and read disturbance require time- and cycle-dependent measurements. Single nominal values establish neither.

E07 Polarization and trapping

Polarization bound charge modifies the screening field and barrier. Trapped charge or ion migration can also change the interfacial electrostatics and show hysteresis. Compare independent polarization measurements with time-resolved resistance after controlled write pulses, and repeat with matched histories and modified ion-blocking/electrode conditions. Neither correlation alone proves exclusivity; probes and leakage can also change results.

E08 Ring isolation

A continuous high-resistance wall across the entire film interrupts every ideal lateral outside-to-inside path even if the interior was never written. Opening a low-resistance gap reconnects a path. The measurement supports a connectivity interpretation under the assumed geometry; it is not a direct oxygen concentration measurement and does not prove microscopic uniformity.

Perovskite switching: Worked solutions (continued)

E09 Humidity and chemical evidence

A charge-balanced model reaction is $H_2O + V_O^{\bullet\bullet} + O_O^{\times} \rightleftharpoons 2OH_O^{\bullet}$: the net relative charge is +2 on both sides. Use matched samples and identical electrical histories, compare dry → intermediate humidity → dry with controlled temperature, and include an untreated control, then compare time dependence and independent chemical probes. Intermediate-humidity behavior in one sample is not a universal trend; changes in contacts, water chemistry and defects may vary by stack.

E10 A partially switched bridge

The outside $190\ \mu\mathrm{m}$ has resistance $(190/200)R_0 = 0.95R_0$. For a uniform film the unchanged half-width band is $(10/200)(20/10)R_0 = 0.1R_0$; its high-resistivity counterpart is $100(0.1R_0) = 10R_0$. Their parallel resistance is $(1/0.1 + 1/10)^{-1}R_0 = 0.0990099R_0$. The total is $1.0490099R_0 \approx 1.04901\ \mathrm{M\Omega}$. Other geometries and contact resistances can produce the same total; the inverse problem is not unique. Two controls are (1) map the written band and use different band widths or an intentional opening to test the geometric path; (2) measure matched untreated devices and four-terminal or separate contact resistance to test contact contributions. Match bias, temperature and write/read history; these tests support but do not by themselves uniquely prove a mechanism.

E11 Oxygen-vacancy fraction and charge balance

The assumed site-based fraction gives $[V_O] = fN_O$, so $\Delta[V_O] = (0.004 - 0.001)(5 \times 10^{22}) = 1.5 \times 10^{20}\ \mathrm{cm^{-3}}$. Each assumed fully ionized vacancy carries relative charge $+2q$; with mobile electrons alone compensating the change, neutrality gives $\Delta n = 2\Delta[V_O] = 3.0 \times 10^{20}\ \mathrm{cm^{-3}}$. Here f is a fraction of oxygen **sites**, not oxygen deficiency per chemical formula unit. Trapped charge, holes, mixed cation valence, immobile dopants and non-equilibrium effects can invalidate this simple balance. Do not separately count the same compensating charge twice.

E12 An interlayer changes the voltage division

For $R_s = 9\,\mathrm{k\Omega}$, the total is $10\,\mathrm{k\Omega}$; $I = 1\,\mathrm{V}/10\,\mathrm{k\Omega} = 0.1\,\mathrm{mA}$ and $V_o = IR_o = 0.1\,\mathrm{V}$. With $R_s = 4\,\mathrm{k\Omega}$, the total is $5\,\mathrm{k\Omega}$, giving $I = 0.2\,\mathrm{mA}$ and $V_o = 0.2\,\mathrm{V}$. Real contacts, interlayers and switching oxides can be nonlinear, history dependent, or evolve chemically; their operating point and thresholds must be measured or modeled with the relevant I–V functions.